Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *MRBENCH* | ***Context:*** *Indian Metropolitan Grade 1–8 Mathematics Teachers (Delhi & Mumbai)*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark is text-only English [Q133, Q134, Q147], matching the deployment's text-based English mobile/enterprise application interface. The target population is fluent in English professionally. No script mismatch, no multimodal mismatch, and adequate infrastructure (urban Tier-1 internet penetration ~80–88%, smartphone access near-universal for professional teachers) [WEB-5, WEB-6, WEB-7] support text-based English signal delivery. Minor concern: no multimodal capture of handwritten student work, which is common in actual Indian classrooms, but the evaluator-role deployment (teachers judging text dialogues) does not require this. The user weighted IF as LOWER priority.

ID	Evidence
Q133	“The prompt template used to generate responses from the seven considered LLMs for both the Bridge and MathDial datasets is shown in Figure 2.” (p.12)
Q134	“The template is adapted from Wang et al. (2024a).” (p.12)
Q147	“In all cases, length is estimated using the number of characters.” (p.14)
WEB-5	Urban Tier-1 internet penetration ~88%, broadband >50% in urban households (https://muftinternet.com/blog/usage-statistics-internet-and-mobile-users-in-india-2025/)
WEB-6	India had 660 million smartphone users as of 2024–2025 (https://www.storyboard18.com/digital/660-million-smartphone-users-16-17-billion-monthly-upi-transactions-power-digital-bharat-report-89731.htm)
WEB-7	National mobile data median speed 94.62 Mbps in early 2024 (https://datareportal.com/reports/digital-2024-india)

Strengths

- Text-only English [Q133, Q134] matches deployment interface exactly
- No script or modality mismatch — both benchmark and deployment are Latin-script English
- Target population has fluent English and reliable mobile infrastructure [WEB-5, WEB-7]
- Length and turn metadata documented per dataset [Q141, Q142, Q147]

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Q141	“It can be observed that the conversation history and response lengths from different LLMs and humans are generally shorter in the Bridge dataset compared to the MathDial dataset.” (p.14)
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions match: text-based English dialogues [Q133, Q134] correspond to deployment's text-based English interface. No image, audio, or specialized capture parameters required.

IF-2: Regional infrastructure supports the format: urban Indian metro internet penetration is high (~80–88%) and smartphone access is near-universal for professional teachers [WEB-5, WEB-6].

IF-3: Domain-specific form differences are minimal. The benchmark does not include images of handwritten work [no quote], which is common in Indian classrooms; however, the evaluator-role deployment (teachers judging text-based AI+human responses) does not require multimodal input.

IF-4: No form mismatches significantly harm external validity for this deployment. Conversation length differences between Bridge and MathDial [Q141, Q142] are documented and do not introduce mismatch with the deployment.

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